BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Year 10 Review and Year 11 Bridge

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Name the IDE facility used to write and change source code.

2. Which description matches an AND gate symbol?

A) Triangle with output circle
B) Curved input; pointed output
C) Flat input; rounded output
D) A decision diamond

3. "Adding 5 to a ticket count always calculates GBP 5 per ticket." Correct this. Explain why.

4. (a) Convert denary 182 to two-digit hexadecimal. (b) Convert hexadecimal B6 to denary. Show your calculation.

Today's objectives

Identify specific gaps using assessment feedback. Improve answers using OCR command words and mark schemes.

Key vocabulary - Read, cover, recall

Command word
Instruction in an exam question, such as state, describe or explain.

Mark scheme
Guidance showing the knowledge and reasoning that earn marks.

Retrieval practice
Recalling knowledge from memory without looking at notes.

Trace table
Table recording variables and other values as an algorithm runs.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Know the two papers	Paper 1 is Computer Systems (sections 1.1-1.6); Paper 2 covers algorithms, programming, robustness, Boolean logic and languages (2.1-2.5). Each paper counts for half the GCSE.
Diagnose the missed mark	Look at the command word and the actual mark scheme. Ask whether the problem was missing knowledge, an incomplete explanation, a tracing error or a calculation mistake.
Improve the response	Write a technical point, then link it to the effect in the situation. 'RAM is fast' is vague; 'RAM holds running instructions and working data so the CPU can access them quickly' is a stronger explanation.
Make a retrieval plan	Choose two named specification points. Test without notes, compare with the mark scheme, correct mistakes, then answer a different question a few days later.

I Do - One worked example

A learner answers 'RAM is fast' to a question asking why RAM is used for currently running programs. Improve this into a full TWO-mark explanation. [2 marks]

1. 'Explain' requires a point and a relevant reason, not only a vague adjective.

2. RAM stores the instructions and data currently being used by the computer [1].

3. The CPU can access that working information quickly when executing the running program [1].

4. Both parts refer to the role of RAM. 'RAM is fast' alone would not secure both marks.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A learner needs a durable portable drive. Explain ONE reason an SSD may be preferable to an HDD. [2 marks]

2. Trace this algorithm: total = 2; total = total + 3; print(total). State the total after both assignments and the output. [3 marks]

3. A message claims to be from a bank and demands an immediate password reset through an unfamiliar link. Identify the likely attack and explain one safe response. [3 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the assignment that totals the three drive capacities.

```
array drivesGB = [64,128,256]
total = 0
for i = 0 to 2
...
next i
print(total)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. A pupil has written a program that stores numbers in an array. Explain why a trace table can help identify a logic error. [3 marks]

Q3. A class uses a Wi-Fi network. Describe TWO actions the school can take to improve network security. [4 marks]

Q4. A school gives a student an assessment with 'state' and 'explain' questions. Describe how the student should respond differently to these two command words. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Writing the name of a feature is always enough to gain two marks for an 'explain' question. ☐ T ☐ F
2. Paper 1 is Computer Systems (sections 1.1-1.6); Paper 2 covers algorithms, programming, robustness, Bool... ☐ T ☐ F
3. Look at the command word and the actual mark scheme ☐ T ☐ F

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Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 The editor. ☐

Q2 C - Flat input; rounded output ☐

Q3 Use total = tickets * 5; addition does not calculate GBP 5 for each ticket. ☐

Q4 (a) B6. (b) 11 x 16 + 6 = 182. ☐

Code First - Model

```
array drivesGB = [64,128,256]
total = 0
for i = 0 to 2
total = total + drivesGB[i]
next i
print(total)
```

Output: 448

We Do - Check the model

1. An SSD has no moving parts [1], so it is less likely to be damaged by knocks when carried [1]. ☐

2. After first line total=2 [1]; after second line total=5 [1]; the output is 5 [1]. ☐

3. Phishing [1]; do not use the link [1] and contact the bank using a known official route to check the request [1]. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 It records relevant variable or array values [1] as the algorithm executes step by step [1], so the point where a value first becomes incorrect can be found [1]. ☐

Q3 Use strong passwords or authentication [1] to reduce unauthorised access [1]. Encrypt the wireless connection [1] to protect data against interception [1]. ☐

Q4 'State' generally requires a concise fact or answer [1]. 'Explain' needs a technical point [1] linked to a reason or effect [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: